THE HANKEL CONTINUED FRACTION OF THE $(n+2)$-SHIFTED q-METALLIC NUMBERS

GUO-NIU HAN

Abstract. For the q-deformed metallic number $\Phi_n(q)$ we prove that the Hankel continued fraction of the $(n+2)$-shifted series $\Phi_n^{(n+2)}$ is eventually periodic, and that the corresponding Hankel determinant sequence $\Delta_j^{(n+2)}$ is $2n(n+1)$-(anti)periodic and takes only the values $-2,-1,0,1,2$. This proves part (1) of a conjecture of Han and Pedon. The determinant values arise as telescoping products of the partial numerators of the Hankel continued fraction. Most of the technically involved and lengthy verifications are carried out by a symbolic computation program, whose code is publicly available on GitHub for independent verification. We emphasize that the program delivers rigorous symbolic proofs — valid for all n and j at once — rather than merely checking finitely many initial terms.

1. Introduction

Morier-Genoud and Ovsienko's theory of q-deformed real numbers [3] attaches to the metallic ratio $\phi_n = \frac{1}{2}\big(n+\sqrt{n^2+4}\big)$ a power series $\Phi_n(q) \in \mathbb{Z}[[q]]$, the *$q$-metallic number*. In [2] the Hankel determinants of Φ_n and of its shifts $\Phi_n^{(\ell)}$ are studied in depth. Write

$$\Delta_j^{(\ell)} := \Delta_j^{(\ell)}(\Phi_n)$$

for the j-th Hankel determinant of the ℓ-shifted series (definitions are recalled in Section 2). For the shifts $0 \le \ell \le n+1$ these determinant sequences are (anti)periodic and take only the values $-1,0,1$; this and the associated Somos–Gale–Robinson dynamics are the main results of [2]. The regime $\ell \ge n+2$ is different, and [2] formulates the following conjecture.

Conjecture 1 (Han–Pedon [2, Conjecture E])**.**

(1) *The sequence $\Delta_j^{(n+2)}$ is $2n(n+1)$-periodic when n is even and $2n(n+1)$-antiperiodic (hence $4n(n+1)$-periodic) when n is odd, and consists of $-2,-1,0,1,2$ only (except if $n=1$: it can only take the values $-1,0,1$).*

(2) *All sequences $\Delta_j^{(\ell)}$ with $\ell \ge n+3$ are unbounded.*

A Hankel determinant sequence confined to the fixed finite set $\{-2,-1,0,1,2\}$ is a rather unusual phenomenon — all the more striking in view of part (2), where a single further shift already makes the determinants unbounded. The author has not encountered such behaviour elsewhere.

The purpose of this paper is to prove part (1) of the conjecture. We obtain it from an explicit description of the *Hankel continued fraction* of $\Phi_n^{(n+2)}$ — a continued fraction expansion of a power series, introduced in [1], whose partial numerators determine all the Hankel determinants of the series; see Section 3 for the precise definition.

Theorem 2. *Let $n \ge 1$.*

(1) *The Hankel continued fraction of $\Phi_n^{(n+2)}$ is eventually periodic.*

Date: July 12, 2026.

2020 *Mathematics Subject Classification.* Primary: 05A30, 11A55, 11C20. Secondary: 11B37.

Key words and phrases. Hankel determinants, Hankel continued fractions, metallic numbers, q-analogues, quadratic irrational series.

(2) *The sequence* $\Delta_j^{(n+2)}$ *is* $2n(n+1)$*-periodic when* n *is even and* $2n(n+1)$*-antiperiodic (hence* $4n(n+1)$*-periodic) when* n *is odd, and consists of* $-2,-1,0,1,2$ *only (except if* $n=1$*: it can only take the values* $-1,0,1$*).*

Part (2) is exactly part (1) of Conjecture 1; for $n=1$ it was first proved by Ovsienko and Pedon [4, Theorem 1.6]. We treat all $n \geq 1$ uniformly, and deduce part (2) from the periodicity in part (1).

Method. To establish Theorem 2 we use the Hankel continued fraction approach developed in [1]; the underlying idea is recalled in Section 3. Its main tool is an algorithm `NextABC` (Section 3.2), which builds the Hankel continued fraction of a quadratic power series one step at a time. Our strategy proceeds as follows.

1. Derive a quadratic equation $A_0 + B_0F + C_0F^2 = 0$ for the shifted series $F = \Phi_n^{(n+2)}$. This is done in Section 2 (Proposition 4).

2. Apply Algorithm `NextABC` to the initial coefficients (A_0, B_0, C_0) derived above. This produces the terms of a sequence of six-tuples

$$\big(A^{(m+1)}, B^{(m+1)}, C^{(m+1)};\, k^{(m)}, a^{(m)}, D^{(m)}\big), \qquad m = 1, 2, 3, \ldots.$$

From these terms we formulate a closed-form, eventually periodic description of the sequence; see Section 3 (with the full pass-tables in the appendix).

3. Prove that the conjectured formulas are correct, i.e. that they satisfy the relations specified in Algorithm `NextABC`. This means checking the three steps of the algorithm: Steps 1 and 2 (the leading data k, a and the polynomial D) are elementary — a least-order term and a short truncation (Section 3.4) — while Step 3 (the three relations for the next triple $A^{(m+1)}, B^{(m+1)}, C^{(m+1)}$) involves the longest expressions. All three are verified symbolically — for all n and j at once — by companion SAGE programs.

4. Construct the Hankel continued fraction of F from the resulting sequence; see Section 3, (19).

5. Finally, compute the Hankel determinants $\Delta_j^{(n+2)}$ from the Hankel continued fraction. This is carried out in Section 4: each $\Delta_j^{(n+2)}$ is a telescoping product of the partial numerators $v_i = -a^{(i)}$, which over one period are constants of the form $\pm 2^e$, so the product collapses to one of $0, \pm 1, \pm 2$ — which is Theorem 2.

On the verification. Algorithm `NextABC` has three steps, each of which we check symbolically, for all n and j at once, after the substitution $q^{2j} \mapsto u$, $q^n \mapsto w$ (turning every entry into a Laurent polynomial in q with coefficients in $\mathbb{Q}(u, w)$). Steps 1 and 2 — the leading data k, a and the truncation defining D in (14) — reduce to comparisons of q-valuations, i.e. to finitely many affine inequalities in (n, j); these are settled by the companion program `certify_steps12.sage`. Step 3 of Algorithm `NextABC` — the relations (15)–(17) for the next triple (A^*, B^*, C^*) — becomes a polynomial identity in q, u, w, valid for all n and j; it is settled by `certify_step3.sage` with no numerical sampling of n. The intermediate expressions are large, exactly as in [1]. All the programs used in the proof, their output logs, and the pass-table data are available at

`https://github.com/GuoniuHan/metallicplus`

Organisation. Section 2 recalls the shifted series $F = \Phi_n^{(n+2)}$ and derives its initial quadratic equation. Section 3 constructs the Hankel continued fraction of F: it introduces the algorithm `NextABC` that generates the tuple sequence, verifies its three steps — Steps 1 and 2 by the valuation argument of Lemma 8, Step 3 as a rational-function identity — and proves that the sequence is eventually periodic (Proposition 9). Section 4 reduces Theorem 2 to the arithmetic bound $E_j \in \{0, 1\}$ and establishes it via the exponent lemma (Lemma 10). The appendices list the explicit pass-tables for one full period: the base cases $n = 1, 2, 3$, the even case $n \geq 4$ (three passes), and the odd case $n \geq 5$ (two passes and a 7-tuple bridge).

2. Setup: the shifted series and its quadratic equation

Throughout, $n \geq 1$ is a fixed integer and

$$Q := q^2 - q + 1, \qquad [n]_q := \frac{1-q^n}{1-q} = 1 + q + \cdots + q^{n-1}.$$

We recall from [2] the q-deformed metallic number $\Phi_n(q)$, the q-deformation of $\phi_n = \frac{n+\sqrt{n^2+4}}{2}$ in the sense of Morier-Genoud and Ovsienko. It is a power series $\Phi_n(q) = \sum_{i\geq 0} f_i q^i \in \mathbb{Z}[[q]]$ with $f_0 = 1$, and it is characterised by the quadratic functional equation

$$q\,\Phi_n(q)^2 + \big((1+q^n)(1-q) - q[n]_q\big)\Phi_n(q) = 1. \tag{1}$$

For $\ell \geq 0$, the *ℓ-shift* of Φ_n is the power series obtained by deleting its first ℓ Taylor coefficients and dividing by q^ℓ:

$$\Phi_n^{(\ell)}(q) := \frac{1}{q^\ell}\Big(\Phi_n(q) - \sum_{i=0}^{\ell-1} f_i q^i\Big) = \sum_{i\geq 0} f_{i+\ell}\, q^i. \tag{2}$$

For a power series $f(q) = \sum_{i\geq 0} f_i q^i$, the *Hankel determinant* of order $j \geq 1$ is

$$\Delta_j(f) := \det\big(f_{a+b}\big)_{0\leq a,b\leq j-1}, \tag{3}$$

with the convention $\Delta_0(f) := 1$. The shifted Hankel determinants of Φ_n are the ordinary Hankel determinants of the shifts, namely $\Delta_j^{(\ell)}(\Phi_n) = \Delta_j\big(\Phi_n^{(\ell)}\big)$; this is the sequence $\Delta_j^{(\ell)}$ studied in the conjecture. Our target is the shift $\ell = n+2$, i.e. the series $F := \Phi_n^{(n+2)}$.

2.1. The quadratic equation of the shift.

The starting point of the whole method is that each shift is again a *quadratic* power series, obtained from the previous one by an elementary substitution.

Lemma 3 ([2, Lemma 7.1])**.** *Let $F(q) = f_0 + f_1 q + \cdots$ satisfy $A + BF + CF^2 = 0$ with A, B, C polynomials, $C \neq 0$. Then the shift $F^{(1)}(q) = \dfrac{F(q) - f_0}{q}$ satisfies $A' + B'F^{(1)} + C'\big(F^{(1)}\big)^2 = 0$, where*

$$A' = \frac{1}{q}\big(A + f_0 B + f_0^2 C\big), \qquad B' = B + 2f_0 C, \qquad C' = qC. \tag{4}$$

Proof. Substitute $F = qF^{(1)} + f_0$ into $A + BF + CF^2 = 0$:

$$\big(A + f_0 B + f_0^2 C\big) + q\,(B + 2f_0 C)\,F^{(1)} + q^2 C\,\big(F^{(1)}\big)^2 = 0.$$

The constant term of $A + f_0 B + f_0^2 C$ equals the constant term of the left-hand side of $A + BF + CF^2 = 0$, hence is 0; so the displayed identity is divisible by q, giving (4). □

Iterating Lemma 3 from (1) produces the quadratic equation of every shift. Writing $A + BF + CF^2 = 0$ for (1) in the normalisation $A^{(0)} = -1$, $B^{(0)} = (1+q^n)(1-q) - q[n]_q$, $C^{(0)} = q$, one proves by induction (see [2, Prop. 7.2]) that for $0 \leq \ell \leq n$

$$A^{(\ell)} = \frac{q^{\ell+1} - Q\,(q^n - q^{n-\ell} + 1)}{(q-1)^2}, \quad B^{(\ell)} = \frac{2q^{\ell+1} - Q\,(q^n+1)}{q-1}, \quad C^{(\ell)} = q^{\ell+1}, \tag{5}$$

while at $\ell = n+1$

$$A^{(n+1)} = -q^{n-1}, \qquad B^{(n+1)} = -\frac{Q + (q^2 - 3q + 1)q^n}{q-1}, \qquad C^{(n+1)} = q^{n+2}. \tag{6}$$

The case we need, $\ell = n+2$, is one further step of Lemma 3.

Proposition 4. *For every integer $n \geq 2$, the shifted series $F = \Phi_n^{(n+2)}$ is a quadratic power series with integer coefficients, satisfying*

$$A_0 + B_0\, F + C_0\, F^2 = 0, \tag{7}$$

where

$$A_0 = -q^{n-2}, \qquad B_0 = -\frac{(q^2-3q+1)\,q^n + Q}{q-1}, \qquad C_0 = q^{n+3}. \tag{8}$$

Moreover $B_0, C_0 \in \mathbb{Z}[q]$ with $B_0(0) = 1$ and $C_0(0) = 0$.

Proof. Apply Lemma 3 to (6). The constant term $f_0 = \Phi_n^{(n+1)}(0)$ is determined by evaluating $A^{(n+1)} + B^{(n+1)} f_0 + C^{(n+1)} f_0^2 = 0$ at $q = 0$: since $n \geq 2$ we have $A^{(n+1)}(0) = 0$, $C^{(n+1)}(0) = 0$ and $B^{(n+1)}(0) = 1$, whence $f_0 = 0$. Thus (4) degenerates to

$$A_0 = \frac{1}{q}\, A^{(n+1)} = -q^{n-2}, \qquad B_0 = B^{(n+1)}, \qquad C_0 = q\, C^{(n+1)} = q^{n+3},$$

which is (8). The numerator of B_0 vanishes at $q = 1$ ($(1-3+1)+Q(1) = -1+1 = 0$), so $B_0 \in \mathbb{Z}[q]$; a direct check gives $B_0(0) = 1$, and $C_0(0) = 0$ is clear. □

Remark 5. *For $n = 1$ the same computation runs one step longer. Formula (6) still holds, giving $A^{(2)} = -1$, $B^{(2)} = 1+q-q^2$, $C^{(2)} = q^3$, but now the constant term of $\Phi_1^{(2)}$ is $f_0 = 1$, and Lemma 3 yields for $F = \Phi_1^{(3)}$ the quadratic equation (7) with*

$$A_0 = Q, \qquad B_0 = 2q^3 - q^2 + q + 1, \qquad C_0 = q^4, \tag{9}$$

again with $B_0(0) = 1$ and $C_0(0) = 0$. This tiny case is treated in Appendix A.

2.2. **Where the algorithm starts.** The Hankel continued fraction of the quadratic series $F = \Phi_n^{(n+2)}$ is computed by iterating Algorithm `NextABC` (denoted N, with parameter $\delta = 2$) of [1, Algorithm 3.1]. The algorithm takes the current triple $(A^{(m)}, B^{(m)}, C^{(m)})$ and returns the next one together with the leading data $(k^{(m)}, a^{(m)}, D^{(m)})$ of the emerging H-fraction; we record the leading monomial $K^{(m)} := a^{(m)} q^{k^{(m)}}$ of $A^{(m)}$. Iterating the algorithm yields the H-fraction of F, and hence, by [1, Theorem 2.1], the determinants $\Delta_j^{(n+2)}$ as telescoping products of the $a^{(m)}$.

Proposition 4 anchors this iteration: the pass-1 tuple at $m = 1$ (equivalently $m = 9j+1$ with $j = 0$) is precisely

$$\big(A^{(1)}, B^{(1)}, C^{(1)}, K^{(1)}, D^{(1)}\big) = \Big(-q^{n-2},\ B_0,\ q^{n+3},\ -q^{n-2},\ -\tfrac{Q-q^n}{q-1}\Big), \tag{10}$$

with $A^{(1)} = A_0$, $B^{(1)} = B_0$, $C^{(1)} = C_0$ from (8); here $K^{(1)}$ is the leading monomial of A_0 (valuation $n-2$, coefficient -1) and $D^{(1)} = -(Q-q^n)/(q-1)$ is the polynomial returned by the first application of Algorithm N. All subsequent tuples of the even and odd passes (Appendices D, E) are generated from (10) by Algorithm N.

For $n = 2, 3$ the tuple (10) specialises to $(-1,\ *,\ q^5,\ -1,\ 1)$ and $(-q,\ *,\ q^6,\ -q,\ q^2+1)$ respectively, matching the base cases of Appendices B, C.

3. The Hankel continued fraction of $\Phi_n^{(n+2)}$

3.1. **Hankel continued fractions.** We first recall from [1] the definition and the basic property of Hankel continued fractions.

Definition 6. *For each positive integer δ, a* super continued fraction *associated with δ, called* super δ-fraction *for short, is a continued fraction of the form*

$$F(q) = \cfrac{v_0 q^{k_0}}{1 + u_1(q)q} - \cfrac{v_1 q^{k_0+k_1+\delta}}{1 + u_2(q)q} - \cfrac{v_2 q^{k_1+k_2+\delta}}{1 + u_3(q)q} - \cdots, \tag{11}$$

where the $v_j \neq 0$ are constants, the k_j are nonnegative integers, and the $u_j(q)$ are polynomials of degree less than or equal to $k_{j-1} + \delta - 2$ (by convention, 0 has degree -1).

When $\delta = 1$ (resp. $\delta = 2$) and all $k_j = 0$, the super δ-fraction (11) is the classical S-fraction (resp. J-fraction). A super 2-fraction is called a *Hankel continued fraction*, or *H-fraction* for short.

Theorem 7 ([1, Theorem 2.1])**.** (i) *Let δ be a positive integer. Each super δ-fraction defines a power series, and conversely, for each power series $F(q)$ the super δ-fraction expansion of $F(q)$ exists and is unique.*

(ii) *Let $F(q)$ be a power series whose H-fraction is given by* (11) *with $\delta = 2$. Then all non-vanishing Hankel determinants of $F(q)$ are given by*

$$H_{s_j}\big(F(q)\big) = (-1)^{\epsilon_j}\, v_0^{s_j} v_1^{s_j - s_1} v_2^{s_j - s_2} \cdots v_{j-1}^{s_j - s_{j-1}}, \tag{12}$$

where $\epsilon_j = \sum_{i=0}^{j-1} k_i(k_i+1)/2$ and $s_j = k_0 + k_1 + \cdots + k_{j-1} + j$ for every $j \geq 0$.

In the statement we have kept the notation of [1]: $H_j(F)$ is the Hankel determinant $\Delta_j(F)$ of (3). We write the H-fraction in the standard form

$$\mathcal{H}\big(q;(k_j),(v_j),(u_j)\big) := \cfrac{v_0 q^{k_0}}{1+u_1(q)q} - \cfrac{v_1 q^{k_0+k_1+2}}{1+u_2(q)q} - \cfrac{v_2 q^{k_1+k_2+2}}{1+u_3(q)q} - \cdots . \tag{13}$$

3.2. **The algorithm `NextABC`.** We recall from [1] the reduction underlying the whole computation. Suppose $F(q)$ is a power series satisfying the quadratic equation

$$A(q) + B(q)F(q) + C(q)F(q)^2 = 0.$$

Then $F(q)$ can be expressed in the form

$$F(q) = \frac{-a\, q^k}{D(q) - q^{k+\delta} G(q)},$$

where $G(q)$ is another power series satisfying a transformed quadratic equation

$$A^*(q) + B^*(q)G(q) + C^*(q)G(q)^2 = 0,$$

and the quantities A^*, B^*, C^*, k, a, D can be computed explicitly by means of the following algorithm (here $\delta = 2$).

Algorithm `[NextABC]` (for $\delta = 2$).

Prototype: $(A^*, B^*, C^*;\, k, a, D) = \texttt{NextABC}(A, B, C)$.

Input: $A(q), B(q), C(q) \in \mathbb{Q}[q]$ three polynomials such that $B(0) = 1$, $C(0) = 0$, $C(q) \neq 0$, $A(q) \neq 0$.

Output: $A^*(q), B^*(q), C^*(q) \in \mathbb{Q}[q]$, $k \in \mathbb{N}$, $a \neq 0 \in \mathbb{Q}$, and $D(q) \in \mathbb{Q}[q]$ a polynomial of degree less than or equal to $k+1$ such that $D(0) = 1$.

Step 1 [Define k, a]. Since $A(q) \neq 0$, let $A(q) = aq^k + O(q^{k+1})$ with $a \neq 0$.

Step 2 [Define D]. Define $D(q)$ by

$$\frac{aq^k B}{A} - \frac{aq^k C}{B} = D(q) + O(q^{k+2}), \tag{14}$$

where $D(q)$ is a polynomial of degree less than or equal to $k+1$ such that $D(0) = 1$.

Step 3 [Define A^*, B^*, C^*]. Let

$$A^*(q) = \big(-D^2 A/a + BD\, q^k - Ca\, q^{2k}\big)/q^{2k+2}, \tag{15}$$

$$B^*(q) = 2AD/(a\, q^k) - B, \tag{16}$$

$$C^*(q) = -A\, q^2/a. \tag{17}$$

Remark. In Step 2, if $C = q^2 C'$ then $aq^k B/A = D(q) + O(q^{k+2})$; similarly, if $A = aq^k$ then $B = D(q) + O(q^{k+2})$.

3.3. **The six-tuple sequence.** Starting from the initial triple (A_0, B_0, C_0) of Proposition 4 and setting $(A^{(1)}, B^{(1)}, C^{(1)}) = (A_0, B_0, C_0)$, repeated application of Algorithm $N = \texttt{NextABC}$ produces six-tuples

$$\big(A^{(m+1)}, B^{(m+1)}, C^{(m+1)};\, k^{(m)}, a^{(m)}, D^{(m)}\big) = N\big(A^{(m)}, B^{(m)}, C^{(m)}\big), \qquad m \ge 1, \tag{18}$$

where $k^{(m)} = \operatorname{val} A^{(m)}$, $a^{(m)}$ is the corresponding leading coefficient (so $K^{(m)} = a^{(m)} q^{k^{(m)}}$), and $D^{(m)}$ is the degree $\le k^{(m)} + 1$ polynomial of Step 2 of the algorithm. By [1] this yields the Hankel continued fraction of F,

$$F = \Phi_n^{(n+2)} = \mathcal{H}\big(q; (k_j), (v_j), (u_j)\big), \qquad k_j = k^{(j+1)}, \quad v_j = -a^{(j+1)}, \quad 1 + u_j(q)\, q = D^{(j)} \ \ (j \ge 1). \tag{19}$$

3.4. **Verification of one step of Algorithm N.** We record how a single application of Algorithm N is verified; this is what proves the periodicity below. The six-tuples (18) are given explicitly, in closed form, in the appendix (Appendices A–E); we refer to these as the *pass-tables*, the term "pass" being explained in §3.5. By inspection, every entry $A^{(m)}, B^{(m)}, C^{(m)}$ is a Laurent polynomial in the two quantities q^{2j} and q^n with coefficients in $\mathbb{Q}(q)$ of bounded degree; recall $B^{(m)}(0) = 1$. In the interior of a pass the index satisfies $j \ge 1$ and $n - 2j$ is large, so the monomials q^{2j}, q^{n-2j}, q^n all have q-order ≥ 2 and are totally ordered; in particular $\operatorname{val} C^{(m)} \ge 2$. For a Laurent series $f = \sum_i c_i q^i$ and an integer d, we write $[f]_{\le d} := \sum_{i \le d} c_i q^i$ for its truncation at order d, i.e. the sum of its terms of degree at most d.

Lemma 8 (Steps 1 and 2)**.** *For every tuple in the pass-tables and every admissible n:*

1. *$A^{(m)}$ has a unique term of least q-order, namely the displayed $K^{(m)} = a^{(m)} q^{k^{(m)}}$; thus $k^{(m)} = \operatorname{val} A^{(m)}$ and $a^{(m)}$ is its leading coefficient.*
2. *$D^{(m)} = \big[a^{(m)} q^{k^{(m)}} B^{(m)} / A^{(m)}\big]_{\le k^{(m)}+1}$, the truncation at order $k^{(m)} + 1$; when $A^{(m)}$ is the monomial $a^{(m)} q^{k^{(m)}}$ this is simply $\big[B^{(m)}\big]_{\le k^{(m)}+1}$.*

Proof. Both assertions depend only on the *shape* of $A^{(m)}$ as a Laurent polynomial in the four blocks 1, q^{2j}, q^{n-2j}, q^n. In the pass's regime ($j \ge 1$ and $n - 2j$ large) these blocks have strictly increasing q-order and their $\mathbb{Q}(q)$-coefficients have bounded degree, so the least-order term of $A^{(m)}$ is controlled by which blocks are present. Inspecting the pass-tables, every $A^{(m)}$ — in each generic row, each $j \in \{0, 1\}$ special row, and each bridge tuple — is of exactly one of two types. We treat the types in turn, verifying assertions (1) and (2) — i.e. Steps 1 and 2 of the algorithm — inside each, and illustrating the verification on a representative tuple.

Type (S)*, a single block.* Here $A^{(m)} = P(q)\, \mu$, where $P \in \mathbb{Q}[q]$ with $P(0) \ne 0$ and μ is one of the monomials $q^{2j+c}, q^{n-2j+c}, q^{n+c}$.

Step 1 (assertion (1))*.* Since μ is a single monomial and $P(0) \ne 0$, we have $A^{(m)} = P(0)\, \mu + \big(\text{higher order in } \mu\big)$, so the term of least q-order is unique and equals $P(0)\, \mu$. Hence $k^{(m)} = \operatorname{val}_q \mu$, $a^{(m)} = P(0)$, and $K^{(m)} = P(0)\, q^{\operatorname{val}_q \mu}$, as recorded.

Step 2 (assertion (2))*.* As $\operatorname{val} C^{(m)} \ge 2$, the C-term of (14) contributes only from order $k^{(m)} + 2$ on and drops out of the truncation (the Remark after the algorithm). Since $q^{k^{(m)}} / A^{(m)} = 1/P$,

$$D^{(m)} = \big[\, a^{(m)} q^{k^{(m)}} B^{(m)} / A^{(m)} \,\big]_{\le k^{(m)}+1} = \big[\, a^{(m)} B^{(m)} / P \,\big]_{\le k^{(m)}+1};$$

the blocks of $B^{(m)}$ of q-order $> k^{(m)} + 1$ contribute nothing, and the truncation of the remainder is $D^{(m)}$. When P is constant, so that $A^{(m)} = a^{(m)} q^{k^{(m)}}$ is itself a monomial, this reduces to $D^{(m)} = [B^{(m)}]_{\le k^{(m)}+1}$.

Worked instance. Take $m = 9j + 3$ in even pass 1 with $j \geq 1$; the pass-table (Appendix D, pass 1) records

$$A^{(m)} = 2q^{\,n-2j-3}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}(1+q^n) + 2(q-2)q^n}{q(q-1)\,q^{2j}},$$
$$C^{(m)} = \frac{Qq^2(q-2)\,q^{2j}(1+q^n) + Qq^4\,q^{4j} + q(q-2)^2q^n}{2(q-1)^2\,q^{2j}},$$
$$K^{(m)} = 2q^{\,n-2j-3}, \qquad D^{(m)} = -\frac{Qq^{2j+1} - q^n}{(q^2-q)q^{2j}}.$$

Here $P = 2$ and $\mu = q^{\,n-2j-3}$. *Step 1:* $A^{(m)}$ is already a monomial, so $k^{(m)} = n - 2j - 3$, $a^{(m)} = 2$, $K^{(m)} = 2q^{\,n-2j-3}$. *Step 2:* being a monomial, $A^{(m)}$ gives $D^{(m)} = [B^{(m)}]_{\leq k^{(m)}+1}$ with $k^{(m)} + 1 = n - 2j - 2$. Writing

$$B^{(m)} = -\frac{Q}{q-1} - \frac{Qq^{2j+1} + 2q - 4}{q^2 - q}\,q^{\,n-2j},$$

the second summand has q-order $\geq n - 2j - 1 > k^{(m)} + 1$, so it is discarded by the truncation; hence $D^{(m)} = \big[-Q/(q-1)\big]_{\leq n-2j-2} = -\dfrac{Qq^{2j+1} - q^n}{(q^2-q)q^{2j}}$, the recorded value.

Type (M)*, several blocks.* Here $A^{(m)}$ contains the q-only block, namely a nonzero summand $A_0(q) \in \mathbb{Q}[q]$ free of q^{2j}, q^{n-2j}, q^n, together with at least one of the higher blocks q^{2j}, q^{n-2j}, q^n; throughout the pass-tables such an A_0 has $\operatorname{val}_q A_0 \leq 1$. (No several-block $A^{(m)}$ built solely from higher blocks occurs: whenever $A^{(m)}$ is not a single block, the q-only part is present.)

Step 1 (assertion (1)). The higher blocks q^{2j}, q^{n-2j}, q^n are all $O(q^2)$ because $j \geq 1$ and $n - 2j$ is large, whereas A_0 has q-order ≤ 1. Hence the term of least q-order comes from A_0; it is unique and gives $k^{(m)} = \operatorname{val}_q A_0 \in \{0, 1\}$ with $a^{(m)}$ the corresponding coefficient, matching the recorded $K^{(m)}$.

Step 2 (assertion (2)). Again $\operatorname{val} C^{(m)} \geq 2$, so the C-term of (14) drops, and

$$D^{(m)} = \big[\, a^{(m)} q^{k^{(m)}} B^{(m)}/A^{(m)} \,\big]_{\leq k^{(m)}+1}$$

is a low-order truncation — a polynomial in q of degree ≤ 2, obtained from $A^{(m)}$ and $B^{(m)}$ modulo $q^{\,k^{(m)}+2}$.

Worked instance. Take $m = 9j + 1$ in even pass 1 with $j \geq 1$; the pass-table (Appendix D, pass 1) records

$$A^{(m)} = -\frac{Q(q-2)^2\,q^{4j} + Q(q-2)\,q^{2j}(1+q^n) + q^{n+1}}{(q-1)^2\,q^{2j}},$$
$$B^{(m)} = -\frac{Q\big(2(q-2)\,q^{2j} + q^n + 1\big)}{q-1}, \qquad C^{(m)} = -Q\,q^{2j},$$
$$K^{(m)} = 2, \qquad D^{(m)} = -\tfrac{1}{2}q + 1.$$

Since $j \geq 1$, the denominator q^{2j} turns the numerator's q^{4j} into a q^{2j}, so the higher blocks present in $A^{(m)}$ are q^{2j}, q^{n-2j}, q^n, all $O(q^2)$; its q-only part $A_0 = -Q(q-2)/(q-1)^2$ carries the rest. Reducing modulo q^2 leaves $A^{(m)} \equiv 2 + q$ and $B^{(m)} \equiv 1$, while $C^{(m)} = O(q^2)$. *Step 1:* the least-order term of $A^{(m)}$ is the constant 2, so $k^{(m)} = 0$, $a^{(m)} = 2$ and $K^{(m)} = a^{(m)}q^{k^{(m)}} = 2$. *Step 2:* the C-term drops, and

$$D^{(m)} = \big[\, 2B^{(m)}/A^{(m)} \,\big]_{\leq 1} = \big[\, 2/(2+q) \,\big]_{\leq 1} = 1 - \tfrac{1}{2}q,$$

matching the recorded $K^{(m)}$ and $D^{(m)}$.

Every tuple of the pass-tables is of type (S) or (M), so Steps 1 and 2 hold for all of them. That this holds *uniformly* — for all admissible n and j at once — is certified by the procedure below, which reduces both assertions to finitely many affine inequalities in (n, j). The two worked instances above are this procedure carried out by hand on the tuples $m = 9j + 3$ (type (S)) and $m = 9j + 1$ (type (M)); the companion program `certify_steps12.sage` runs it on every tuple. □

Algorithm `[CertifySteps12]`.

Input: a pass-table tuple $(A, B, C; K, D)$ and a cone $\mathcal{C} = \{\, j \ge j_0,\ n - 2j \ge N \,\}$, with $j_0, N \in \mathbb{N}$.

Output: TRUE if assertions (1) and (2) of Lemma 8 hold at every $(n, j) \in \mathcal{C}$.

Notation. Write each expression as a finite sum of *blocks* $X = \sum_{b,c} P_{b,c}(q)\, u^b w^c$, where $u = q^{2j}$, $w = q^n$ and $P_{b,c} \in \mathbb{Q}(q)$. For the pass-table entries A, B, C, D the index (b, c) takes only the four values $(0,0), (1,0), (-1,1), (0,1)$ — the blocks $1, q^{2j}, q^{n-2j}, q^n$ — but the same decomposition is applied in Step 2 to the auxiliary polynomial M, whose blocks (being products of these) range over a larger, still finite, set. Under $u = q^{2j}, w = q^n$ the block (b, c) has q-order

$$\operatorname{ord}(b, c) = \operatorname{val}_q P_{b,c} + 2jb + nc, \tag{20}$$

an affine form in (j, n). For an affine form $g = \alpha + \beta j + \gamma n$ and a target $\tau \in \mathbb{Z}$, write $g \succeq_{\mathcal{C}} \tau$ for the predicate "$g(j, n) \ge \tau$ for all $(n, j) \in \mathcal{C}$". Substituting $j = j_0 + s$ and $n - 2j = N + t$ with $s, t \ge 0$ gives the finite test

$$g \succeq_{\mathcal{C}} \tau \iff \beta + 2\gamma \ge 0, \quad \gamma \ge 0, \quad \alpha + \beta j_0 + \gamma(2j_0 + N) \ge \tau, \tag{21}$$

a check on the three coefficients only.

Step 1 [least-order term]. Write $K = a\, q^{k_0} u^{b_0} w^{c_0}$. Verify that block (b_0, c_0) occurs in A with $\operatorname{val}_q P_{b_0,c_0} = k_0$ and leading coefficient a, and that $\operatorname{ord}(b, c) - \operatorname{ord}(b_0, c_0) \succeq_{\mathcal{C}} 1$ for every other block (b, c) of A. This certifies that K is the unique term of least q-order, so $k^{(m)} = \operatorname{val} A = k_0 + 2jb_0 + nc_0$ with leading coefficient a — assertion (1).

Step 2 [truncation D]. Verify that B has block $(0, 0)$ of q-order 0 and every other block of B strictly above it, so $\operatorname{val} B = 0$. Form

$$M := a\, q^k B^2 - a\, q^k C\, A - D\, A\, B, \tag{22}$$

and verify $\operatorname{ord}(b, c) \succeq_{\mathcal{C}} 2k + 2$ for every block of M. Finally, let δ be the affine q-degree of the denominator of D and verify, for every monomial $q^i u^b w^c$ of its numerator, $(k+1) + \delta - \operatorname{ord}(q^i u^b w^c) \succeq_{\mathcal{C}} 0$. Return TRUE if all checks pass.

Correctness. Step 1 is assertion (1). In Step 2, the block bounds on M give $\operatorname{val}_q M \ge 2k + 2$, hence $\operatorname{val}_q\big(aq^k B/A - aq^k C/B - D\big) = \operatorname{val}_q M - \operatorname{val}_q(AB) \ge k + 2$ (using $\operatorname{val} A = k$, $\operatorname{val} B = 0$); the C-term is harmless here as $\operatorname{val} C \ge 2$ forces $\operatorname{val}_q(aq^k C/B) \ge k + 2$. The last check gives $\deg_q D \le k + 1$. A Laurent polynomial that agrees with $aq^k B/A$ modulo q^{k+2} and has q-degree $\le k + 1$ is exactly $[aq^k B/A]_{\le k+1}$; thus $D = D^{(m)}$ — assertion (2).

The program `certify_steps12.sage` runs `CertifySteps12` on every tuple of the pass-tables, walking the tables in order with their case conditions (Type (S)/(M) and $j \in \{0, 1\}$ / generic / bridge) and, for each, taking the smallest N for which it returns TRUE. It additionally checks *coverage*: each tuple is used only within its certified cone, the least $d = n - 2j$ at which a tuple occurs being a constant read off from the pass boundary $B_p = \frac{9}{2}n + O(1)$. All tuples pass, which proves Lemma 8.

Step 3 of the algorithm — the three relations (15)–(17) defining $(A^{(m+1)}, B^{(m+1)}, C^{(m+1)})$ — is, by contrast, a rational-function identity in q that requires no ordering of exponents. Writing $w = q^n$ and $u = q^{2j}$ (so $q^{4j} = u^2$, $q^{n-2j} = w/u$) turns each entry into a Laurent polynomial in q with coefficients in $\mathbb{Q}(u, w)$, and (15)–(17) become polynomial identities in q, u, w — valid for all n and j simultaneously. All the transitions occurring below (both parities, the generic and special

rows, and the odd bridge) are verified in this form by the companion program `certify_step3.sage` (which reads the pass-table data from `passtables.sage`), each additionally cross-checked against a direct run of Algorithm N at a concrete (n, j).

3.5. **Periodicity.** *Passes.* It is convenient to group the index m into blocks of nine: write $m = 9j + r$ with $r \in \{1, \dots, 9\}$, so that r is the position within a block and $j \geq 0$ is the block number. As j increases by 1, the index m increases by 9. Inspecting the tables, one finds that a single family of nine closed-form expressions — one for each residue r, depending on j only through the monomials q^{2j} and q^{n-2j} — describes the tuples $(A^{(m)}, B^{(m)}, C^{(m)}; k^{(m)}, a^{(m)}, D^{(m)})$ over a whole range of consecutive blocks j. Such a maximal run of blocks governed by one family of nine formulas is what we call a *pass*. A pass ends at the block where the exponents $2j$ and $n-2j$ change relative order — concretely, where $n-2j$ becomes small; at that "boundary" the leading terms rearrange, the nine formulas cease to apply, and a new pass, with a different family of nine formulas, takes over. Since j runs from 0 to roughly $n/2$, each pass spans about $\frac{9}{2}n$ tuples. (For the small cases $n = 1, 2, 3$ there is a single short period, which we simply list.)

The heart of the matter is that this process *closes up*: after finitely many passes the sequence returns to an earlier one, so that only finitely many distinct passes occur and the tuple sequence (18) is eventually periodic. We stress that this is not clear a priori. The tuples range over an infinite set, so — unlike the situation over a finite field, where every orbit is eventually periodic by pigeonhole [1] — there is no general reason for such a return; it is a fact established by carrying out Algorithm N explicitly and observing that the passes close into a cycle. The resulting closed form depends only on the parity of n: for even n, after the transient pass 1 the passes 2 and 3 repeat (pass 4 being again pass 2), giving period $9n - 10$; for odd n, only the single initial tuple is transient, and the sequence then runs through passes 1 and 2 and a short 7-tuple *bridge*, which returns to pass 1, giving period $9n - 13$.

Proposition 9. *Let $n \geq 1$.*

1. (Base cases.) *For $n = 1$ the tuples $(A^{(m)}, \dots)$ are periodic in m with period 2 from $m = 2$ on; for $n = 2$ they are periodic with period 9 from $m = 2$ on; for $n = 3$ with period 14 from $m = 2$ on. The explicit tuples are listed in Appendices* A–C.
2. (Even $n \geq 4$.) *The tuples are described in closed form by the three "passes" of Appendix* D*; after a transient the sequence is periodic, and pass 4 coincides with pass 2, so the tuple sequence closes into a cycle.*
3. (Odd $n \geq 5$.) *The orbit from the initial tuple has pre-period 1 and period $9n - 13$. Passes 1 and 2 of Appendix* E *account for $9n - 20$ of these tuples; the remaining 7 tuples per period form an explicit* bridge*, given in closed form in Appendix E (subsection E.3), after which the orbit returns to pass 1 (index $m = 2$). Its transitions are verified exactly as in §3.4.*

In every case the partial numerators $v_j = -a^{(j+1)}$ are nonzero constants of the form $\pm 2^e$ with $e \in \{-2, -1, 0, 1, 2\}$, and only finitely many distinct tuples occur.

Proof. Each clause asserts that a displayed six-tuple is the image under Algorithm N of the preceding one, for every index m in the relevant range, together with the "closing" identities (even n: pass 4 = pass 2; odd n: the bridge returns to pass 1 at $m = 2$). By Lemma 8 the leading data $k^{(m)}, a^{(m)}$ and the polynomial $D^{(m)}$ (Steps 1 and 2) are as displayed, for all n; and Step 3, the relations (15)–(17) for $(A^{(m+1)}, B^{(m+1)}, C^{(m+1)})$, holds identically in q, u, w by the symbolic verification described above (`certify_step3.sage`). Thus each transition is valid for all n. The closing identities and the base cases $n = 1, 2, 3$ are direct equalities of the displayed tables. Finally, the partial numerators $v_j = -a^{(j+1)}$ and the bound $e \in \{-2, \dots, 2\}$ are read off from the K column of the tables. □

Proposition 9 reduces Theorem 2 to a finite computation: over one period the data (k_j, v_j) take finitely many values, and we must show that the resulting Hankel determinants (19) lie in $\{-2,-1,0,1,2\}$. This is carried out in the next section.

4. Proof of Theorem 2

We apply Theorem 7(ii) to the H-fraction (19) of $F = \Phi_n^{(n+2)}$. With the data (k_j, v_j) of (19), formula (12) says that the only nonzero Hankel determinants of F are the $\Delta_{s_j}(F)$ at the indices

$$s_0 = 0, \qquad s_j = k_0 + k_1 + \cdots + k_{j-1} + j \quad (j \geq 1), \tag{23}$$

and there

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j}\, v_0^{s_j}\, v_1^{s_j - s_1} \cdots v_{j-1}^{s_j - s_{j-1}}, \qquad \epsilon_j = \sum_{i=0}^{j-1} \binom{k_i + 1}{2}. \tag{24}$$

All other Hankel determinants of F vanish; these account for the value 0 in Theorem 2. It therefore remains to show that each product (24) equals ± 1 or ± 2.

4.1. **Reduction to an exponent count.** By Proposition 9 every partial numerator is of the form $v_i = \pm 2^{e_i}$ with $e_i \in \mathbb{Z}$. Writing $\sigma_i = \operatorname{sgn} v_i$, the product (24) is

$$\Delta_{s_j}(F) = (-1)^{\epsilon_j} \Big(\textstyle\prod_{i<j} \sigma_i^{\,s_j - s_i} \Big)\, 2^{E_j}, \qquad E_j := \textstyle\sum_{i=0}^{j-1} e_i\,(s_j - s_i). \tag{25}$$

Thus $\Delta_{s_j}(F) \in \{-2,-1,0,1,2\}$ if and only if $E_j \in \{0,1\}$ for every j (the sign is automatically $\pm$). So Theorem 2 is equivalent to the purely arithmetic statement

$$E_j \in \{0,1\} \qquad \text{for all } j \geq 0. \tag{26}$$

4.2. **The exponent lemma.** The bound (26) rests on a remarkable alignment between the k-values and the exponents e_i. Recall $s_{i+1} - s_i = k_i + 1$; write

$$d_i := k_i + 1 \geq 1, \qquad T_i := \sum_{l=0}^{i} e_l, \tag{27}$$

so that T_i is the i-th partial sum of the exponent sequence. Reversing the order of summation in (25),

$$E_j = \sum_{i=0}^{j-1} e_i \sum_{l=i}^{j-1} d_l = \sum_{l=0}^{j-1} d_l \sum_{i=0}^{l} e_i = \sum_{l=0}^{j-1} d_l\, T_l. \tag{28}$$

Lemma 10. *For every $n \geq 1$ the periodic data (k_i, e_i) of the H-fraction of $\Phi_n^{(n+2)}$ satisfies:*

(F1) *$T_i \in \{-1,0,1\}$ for all i;*
(F2) *$k_i = 0$ whenever $T_i \neq 0$;*
(F3) *the partial sums $\displaystyle\sum_{l=0}^{j-1} T_l$ lie in $\{0,1\}$ for all $j \geq 0$.*

Consequently $E_j \in \{0,1\}$ for all j, so (26) holds.

Proof. For the consequence: by (F2), $d_i T_i = T_i$ for every i (if $T_i = 0$ both sides vanish; if $T_i \neq 0$ then $k_i = 0$, i.e. $d_i = 1$). Hence (28) collapses to $E_j = \sum_{l<j} T_l$, which lies in $\{0,1\}$ by (F3).

It remains to prove (F1)–(F3). For $n = 1$ all $a^{(m)} = \pm 1$ (Appendix A), so $e_i \equiv 0$, $T_i \equiv 0$, and (F1)–(F3) are trivial.

For $n \geq 2$ we read the data off Proposition 9. Away from the pass junctions (the pre-period and the special rows $j \in \{0,1\}$, treated below), the leading coefficient $a^{(m)}$ and the valuation $k^{(m)} = \operatorname{val} A^{(m)}$ depend only on the residue $r \in \{1,\dots,9\}$ of m inside a pass, not on j:

r	1	2	3	4	5	6	7	8	9
$a^{(m)}$, pass 1	2	$-\frac{1}{4}$	2	-1	1	1	1	1	-1
pass 2	1	1	1	-1	2	$-\frac{1}{4}$	2	-1	1
pass 3	2	-1	1	1	1	1	-1	2	$-\frac{1}{4}$
odd pass 2	-1	1	1	1	1	-1	2	$-\frac{1}{4}$	2

(odd pass 1 equals even pass 1); the matching valuations are

r	1	2	3	4	5	6	7	8	9
$k^{(m)}$, pass 1	0	0	$n-2j-3$	0	$2j-1$	1	$n-2j-4$	0	$2j$
pass 2	1	$n-2j-3$	0	$2j-1$	0	0	$n-2j-4$	0	$2j$
pass 3	$n-2j-1$	0	$2j-3$	1	$n-2j-2$	0	$2j-2$	0	0
odd pass 2	0	$2j-1$	1	$n-2j-4$	0	$2j$	0	0	$n-2j-5$

and, with $e_i = v_2(a^{(m)})$ (so $2 \mapsto 1$, $-\frac{1}{4} \mapsto -2$, $\pm 1 \mapsto 0$),

r	1	2	3	4	5	6	7	8	9
e_i, pass 1	1	-2	1	0	0	0	0	0	0
pass 2	0	0	0	0	1	-2	1	0	0
pass 3	1	0	0	0	0	0	0	1	-2
odd pass 2	0	0	0	0	0	0	1	-2	1

Each row of the last table is the *same* block $(1,-2,1)$, occupying three consecutive residues (cyclically: $r = 8,9,1$ in pass 3) and 0 elsewhere; its entries carry $a^{(m)} = 2, -\frac{1}{4}, 2$ with, from the k-table, $k^{(m)} = 0,0$ and a positive value. Hence along one 9-cycle the running total $T_i = \sum_{l \leq i} e_l$ reads $0,\dots,0,1,-1,0,\dots,0$: it enters the block at 0, visits 1 and -1, and returns to 0. So within a pass $T_i \in \{-1,0,1\}$ (F1); the only residues with $T_i \neq 0$ are the two carrying $a = 2, -\frac{1}{4}$, both with $k^{(m)} = 0$ (F2); and the partial sums $E_j = \sum_{l<j} T_l$ stay in $\{0,1\}$ (F3).

Junctions. The orbit strings these passes together through the pre-period and the special rows $j \in \{0,1\}$ — a finite set of tuples, the same for all n. Cutting the whole orbit at the points where $T = 0$, one finds only three excursion shapes: the block $(1,-2,1)$ above and, at the even pass junctions, the two blocks $(1,-2,2,-1)$ and $(-1,2,-2,1)$ (from $a^{(m)} = 2, -\frac{1}{4}, 4, \frac{1}{2}$ and $\frac{1}{2}, 4, -\frac{1}{4}, 2$), with running totals $(1,-1,1,0)$ and $(-1,1,-1,0)$. Each of the three stays in $\{-1,0,1\}$, returns to 0, and has $k^{(m)} = 0$ at every residue where $T \neq 0$; so (F1)–(F3) persist across the junctions. (For odd n every excursion is already $(1,-2,1)$, the bridge included.) The placement of the junctions is what makes this work: were a pass glued so that a $+1$ — an $a^{(m)} = 2$ — landed on a residue already at $T = 1$, the total would reach 2 and (F1) would fail. The tabulated junctions all avoid this. □

Proof of Theorem 2. Part (1) is Proposition 9: for every n the six-tuple sequence, hence the H-fraction data (k_j, v_j, u_j) of (19), is eventually periodic.

Part (2), the value set. For $n \geq 2$, combining (25) with Lemma 10, every nonzero $\Delta_j^{(n+2)}$ equals $\pm 2^{E_j}$ with $E_j \in \{0,1\}$, hence lies in $\{-2,-1,1,2\}$, while the remaining indices give $\Delta_j^{(n+2)} = 0$; thus $\Delta_j^{(n+2)} \in \{-2,-1,0,1,2\}$. For $n = 1$ the H-fraction of $\Phi_1^{(3)}$ has all partial numerators $v_i = \pm 1$ (Appendix A), i.e. $e_i \equiv 0$, so (25) gives $E_j \equiv 0$ and $\Delta_j^{(3)} \in \{-1,0,1\}$, recovering [4, Theorem 1.6].

Part (2), the (anti)periodicity. By part (1) the data (k_i, v_i) is periodic in i, with some period P, from $i = 1$ on. Moreover, the tables show that the tuple at index $1+P$ is $\left(-A^{(1)}, B^{(1)}, -C^{(1)}\right)$ — the initial tuple up to sign (compare the initial tuple with even pass 3, $m = 8$, $j = 0$; the bridge

tuple b7; and the last listed tuple for $n = 1, 2, 3$) — so that $k_{i+P} = k_i$ and $|v_{i+P}| = |v_i|$ hold for *all* $i \geq 0$. Summing the tables gives $\sum_i (k_i+1) = 2n(n+1)$ over one period; hence $s_{j+P} = s_j + 2n(n+1)$ for all $j \geq 0$, so an index N is of the form s_j if and only if $N + 2n(n+1)$ is, and the vanishing pattern of $\Delta^{(n+2)}$ repeats with period $2n(n+1)$. At the surviving indices we compare $\Delta_{s_{j+P}}$ with Δ_{s_j} through (25). The 2-parts agree, $E_{j+P} = E_j$: indeed $E_j = \sum_{l<j} T_l$, and the sum of the e_i (hence of the T_l) over any P consecutive indices is independent of the starting point and must vanish, since otherwise the bounds (F1), (F3) of Lemma 10 would fail for large j. The ratio of the signs is a finite product of the data $(\sigma_i,\ s_i \bmod 2,\ k_i)$ over one period, read off from the tables; evaluating it — exactly as in the proof of the analogous statement for the shifts $0 \leq \ell \leq n+1$ in [2] — gives the constant value $+1$ for n even and -1 for n odd, in accordance with the parity of $\sum_i \binom{k_i+1}{2}$ over one period. Hence $\Delta^{(n+2)}_{N+2n(n+1)} = (-1)^n \Delta^{(n+2)}_N$ for all $N \geq 0$: the sequence is $2n(n+1)$-periodic for n even and $2n(n+1)$-antiperiodic for n odd. Together with the value set, this is part (2). □

Appendix A. Base case $n = 1$

The initial triple is given by Remark 5; write $B := 2q^3 - q^2 + q + 1$, so $B^{(m)} = B$ for all m. The orbit has pre-period 1 and period 2 (from $m = 2$), and $m = 4$ maps back to $m = 2$; the three transitions are identities between concrete polynomials in q and are checked directly. Since Algorithm N runs verbatim at a numerical n, the tuples for $n = 1, 2, 3$ are not guessed but obtained by iterating it from the initial triple; this is done in `verify_base.sage`. Note that all $a^{(m)} = \pm 1$ and that $\sum_m \big(k^{(m)} + 1\big) = 3 + 1 = 4 = 2n(n+1)$ over one period.

$m = 1$:

$$A^{(m)} = Q, \qquad C^{(m)} = q^4, \qquad K^{(m)} = 1, \qquad D^{(m)} = 2\,q + 1$$

$m = 2$:

$$A^{(m)} = -q^2, \qquad C^{(m)} = -Qq^2, \qquad K^{(m)} = -q^2, \qquad D^{(m)} = B$$

$m = 3$:

$$A^{(m)} = -Q, \qquad C^{(m)} = -q^4, \qquad K^{(m)} = -1, \qquad D^{(m)} = 2\,q + 1$$

Appendix B. Base case $n = 2$

The 10 tuples $\big(A^{(m)}, B^{(m)}, C^{(m)}; K^{(m)}, D^{(m)}\big)$, $m = 1, \ldots, 10$; the period is 9 (from $m = 2$), and $m = 10$ maps back to $m = 2$. Computed in `verify_base.sage`. $m = 1$:

$$A^{(m)} = -1, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = q^5, \qquad K^{(m)} = -1, \qquad D^{(m)} = 1$$

$m = 2$:

$$A^{(m)} = q^3 - q + 2, \qquad B^{(m)} = q^3 - 2\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = 2$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A^{(m)} = -\frac{1}{8}\,q^3 - \frac{3}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2$$
$$K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = \frac{3}{2}\,q + 1$$

$m = 4$:

$$A^{(m)} = 2\,q^3 + 2\,q^2 + 6\,q + 4, \qquad B^{(m)} = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2$$

$$K^{(m)} = 4, \qquad D^{(m)} = -\frac{3}{2}\,q + 1$$

$m = 5$:

$$A^{(m)} = -\frac{1}{8}\,q^3 - \frac{1}{8}\,q^2 - \frac{3}{8}\,q - \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{1}{2}\,q^3 + \frac{5}{2}\,q^2 + 1$$
$$C^{(m)} = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{3}{2}\,q + 1$$

$m = 6$:

$$A^{(m)} = -2\,q^3 - 6\,q + 4, \qquad B^{(m)} = -2\,q^4 - q^3 - 6\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 - \frac{3}{2}\,q^3 - q^2$$
$$K^{(m)} = 4, \qquad D^{(m)} = \frac{3}{2}\,q + 1$$

$m = 7$:

$$A^{(m)} = \frac{1}{8}\,q^3 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^4 + \frac{3}{2}\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^5 + \frac{3}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 8$:

$$A^{(m)} = 2, \qquad B^{(m)} = q^3 - 2\,q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^5 + \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = 2, \qquad D^{(m)} = 1$$

$m = 9$:

$$A^{(m)} = q^3, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = q^3$$
$$D^{(m)} = -q^3 + 2\,q^2 + 1$$

$m = 10$:

$$A^{(m)} = 1, \qquad B^{(m)} = -q^3 + 2\,q^2 + 1, \qquad C^{(m)} = -q^5, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

Appendix C. Base case $n = 3$

The 15 tuples $m = 1, \ldots, 15$; period 14 (from $m = 2$), $m = 15$ maps back to $m = 2$. Computed in `verify_base.sage`. $m = 1$:

$$A^{(m)} = -q, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = q^6, \qquad K^{(m)} = -q, \qquad D^{(m)} = q^2 + 1$$

$m = 2$:

$$A^{(m)} = q^4 - q^3 + 2\,q^2 - q + 2, \qquad B^{(m)} = q^4 - 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^3, \qquad K^{(m)} = 2$$
$$D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 3$:

$$A^{(m)} = -\frac{1}{8}\,q^4 + \frac{1}{8}\,q^3 - \frac{1}{4}\,q^2 + \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C^{(m)} = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 4$:

$$A^{(m)} = 2, \qquad B^{(m)} = -q^4 - 3\,q^2 + 1, \qquad C^{(m)} = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = 2$$
$$D^{(m)} = 1$$

$m = 5$:

$$A^{(m)} = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B^{(m)} = q^4 + 3\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = -1$$
$$D^{(m)} = q + 1$$

$m = 6$:

$$A^{(m)} = -q^4 - q^2 - q, \qquad B^{(m)} = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1$$
$$C^{(m)} = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K^{(m)} = -q, \qquad D^{(m)} = 2\,q^2 - q + 1$$

$m = 7$:

$$A^{(m)} = -q^4 + q^3 - q^2, \qquad B^{(m)} = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 - q^4 - q^3$$
$$K^{(m)} = -q^2, \qquad D^{(m)} = 2\,q^3 + q^2 + q + 1$$

$m = 8$:

$$A^{(m)} = -q^4 - q^2 - q, \qquad B^{(m)} = 2\,q^5 - q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 + q^5 - q^4$$
$$K^{(m)} = -q, \qquad D^{(m)} = 2\,q^2 - q + 1$$

$m = 9$:

$$A^{(m)} = -q^4 + q^3 - 3\,q^2 + q - 1, \qquad B^{(m)} = 2\,q^5 - q^4 + 4\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6 - q^4 - q^3$$
$$K^{(m)} = -1, \qquad D^{(m)} = q + 1$$

$m = 10$:

$$A^{(m)} = -1, \qquad B^{(m)} = q^4 + 3\,q^2 + 1, \qquad C^{(m)} = -q^6 + q^5 - 3\,q^4 + q^3 - q^2, \qquad K^{(m)} = -1$$
$$D^{(m)} = 1$$

$m = 11$:

$$A^{(m)} = -q^4 + q^3 - 2\,q^2 + q + 2, \qquad B^{(m)} = -q^4 - 3\,q^2 + 1, \qquad C^{(m)} = -q^2, \qquad K^{(m)} = 2$$
$$D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 12$:

$$A^{(m)} = \frac{1}{8}\,q^4 - \frac{1}{8}\,q^3 + \frac{1}{4}\,q^2 - \frac{1}{8}\,q + \frac{1}{4}, \qquad B^{(m)} = \frac{1}{2}\,q^5 - \frac{1}{2}\,q^4 + 2\,q^3 + \frac{1}{2}\,q^2 + 1$$
$$C^{(m)} = \frac{1}{2}\,q^6 - \frac{1}{2}\,q^5 + q^4 - \frac{1}{2}\,q^3 - q^2, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 13$:

$$A^{(m)} = 2\,q, \qquad B^{(m)} = q^4 - 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -\frac{1}{2}\,q^6 + \frac{1}{2}\,q^5 - q^4 + \frac{1}{2}\,q^3 - q^2$$
$$K^{(m)} = 2\,q, \qquad D^{(m)} = q^2 + 1$$

$m = 14$:

$$A^{(m)} = q^4, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^3, \qquad K^{(m)} = q^4$$
$$D^{(m)} = -q^4 + 2\,q^3 + q^2 + 1$$

$m = 15$:

$$A^{(m)} = q, \qquad B^{(m)} = -q^4 + 2\,q^3 + q^2 + 1, \qquad C^{(m)} = -q^6, \qquad K^{(m)} = q, \qquad D^{(m)} = q^2 + 1$$

Appendix D. Even $n \geq 4$: the three passes

The orbit has period $9n-10$. Passes 1–3 below describe it; the junction tuples are shared (pass $1[B_1]$ = pass $2[m{=}1]$, pass $2[B_2]$ = pass $3[m{=}1]$), and pass 4 coincides with pass 2, closing the cycle. Boundary indices $B_1 = \frac{9}{2}n-7$, $B_2 = \frac{9}{2}n-8$, $B_3 = \frac{9}{2}n$. The transitions are proved for all even n as described in §3.4, each additionally cross-checked against a direct run of Algorithm N at one concrete (n, j) (the guard built into `passtables.sage`).

D.1. **Pass 1 (transient head).** $m=1$, $m=9j+1$, $j=0$:

$$A^{(m)} = -q^{n-2}, \qquad B^{(m)} = -\frac{\left(q^2-3\,q+1\right)q^n+Q}{q-1}, \qquad C^{(m)} = q^{n+3}, \qquad K^{(m)} = -q^{n-2}$$

$$D^{(m)} = -\frac{Q-q^n}{q-1}$$

$m=9j+1$, $j \notin \{0\}$:

$$A^{(m)} = -\frac{Q(q-2)^2q^{4j}+Q(q-2)q^{2j}\,(1+q^n)+q^{n+1}}{(q-1)^2q^{2j}}, \qquad B^{(m)} = -\frac{Q\left(2(q-2)q^{2j}+q^n+1\right)}{q-1}$$

$$C^{(m)} = -Qq^{2\,j}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m=2$, $m=9j+2$, $j=0$:

$$A^{(m)} = \frac{\left(q^2-2q+2\right)q^n+Q(q-2)}{q-1}, \qquad B^{(m)} = \frac{\left(q^2-3\,q+3\right)q^n-Q}{q-1}, \qquad C^{(m)} = -q^n$$

$$K^{(m)} = 2, \qquad D^{(m)} = \frac{1}{2}\,q+1$$

$m=9j+2$, $j \notin \{0\}$:

$$A^{(m)} = \frac{Qq(q-2)q^{2j}\,(1+q^n)+Qq^3q^{4j}+(q-2)^2q^n}{8q(q-1)^2q^{2j}}$$

$$B^{(m)} = \frac{Qq^2(q-2)q^{4j}+Q\left(q^2-2q+2\right)q^{2j}\,(1+q^n)+q(q-2)q^n}{2(q-1)^2q^{2j}}$$

$$C^{(m)} = \frac{Qq^2(q-2)^2q^{4j}+Qq^2(q-2)q^{2j}\,(1+q^n)+q^3q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m=3$, $m=9j+3$, $j=0$:

$$A^{(m)} = -\frac{Qq(q+2)+(q-2)\left(q^2+2\right)q^n}{8q(q-1)}, \qquad B^{(m)} = \frac{Q\left(q^2-2\right)+\left(q^3-2q^2+4q-2\right)q^n}{2(q-1)}$$

$$C^{(m)} = -\frac{q^2\left(Q(q-2)+\left(q^2-2q+2\right)q^n\right)}{2(q-1)}, \qquad K^{(m)} = \frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m=9j+3$, $j \notin \{0\}$:

$$A^{(m)} = 2\,q^{-2\,j+n-3}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}\,(1+q^n)+2(q-2)q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = \frac{Qq^2(q-2)q^{2j}\,(1+q^n)+Qq^4q^{4j}+q(q-2)^2q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = 2\,q^{-2\,j+n-3}$$

$$D^{(m)} = -\frac{Qq^{2\,j+1}-q^n}{(q^2-q)q^{2\,j}}$$

$m=4$, $m=9j+4$, $j=0$:

$$A^{(m)} = 2\,q^{n-3}, \qquad B^{(m)} = -\frac{\left(q^3 - q^2 + 3\,q - 4\right)q^n + Qq}{q^2 - q}$$

$$C^{(m)} = \frac{q\left(Qq(q+2) + (q-2)\left(q^2+2\right)q^n\right)}{2(q-1)}, \qquad K^{(m)} = 2\,q^{n-3}, \qquad D^{(m)} = -\frac{Qq - q^n}{q^2 - q}$$

$m = 9j + 4$, $j \notin \{0\}$:

$$A^{(m)} = \frac{Qq^2(q-1)q^{4j} + Qq\,q^{2j}\,(1-q^n) - (q-1)q^n}{q(q-1)q^{2j}}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}\,(1-q^n) - 2(q-1)q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = -q^{-2\,j+n-1}, \qquad K^{(m)} = -1, \qquad D^{(m)} = 1$$

$m = 5$, $m = 9j + 5$, $j = 0$:

$$A^{(m)} = \frac{Q^2q - \left(q^3 - q^2 + 2\,q - 1\right)q^n}{q^2 - q}, \qquad B^{(m)} = \frac{\left(q^3 - q^2 + 3\,q - 2\right)q^n - Qq}{q^2 - q}$$

$$C^{(m)} = -q^{n-1}, \qquad K^{(m)} = -1, \qquad D^{(m)} = q + 1$$

$m = 9j + 5$, $j \notin \{0\}$:

$$A^{(m)} = Qq^{2\,j-1}, \qquad B^{(m)} = -\frac{Q\left(2q(q-1)q^{2j} - q^n + 1\right)}{q-1}$$

$$C^{(m)} = \frac{Qq^3(q-1)q^{4j} + Qq^2q^{2j}\,(1-q^n) - q(q-1)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = q^{2\,j-1}, \qquad D^{(m)} = \frac{q^{2\,j+1} - 1}{q-1}$$

$m = 6$, $m = 9j + 6$, $j = 0$:

$$A^{(m)} = \frac{Qq(q+1) - \left(q^2+1\right)q^n}{q-1}, \qquad B^{(m)} = -\frac{Q\left(2q^3+1\right) - \left(2q^3 - q^2 + 3q - 1\right)q^n}{q-1}$$

$$C^{(m)} = \frac{q\left(Q^2q - \left(q^3 - q^2 + 2q - 1\right)q^n\right)}{q-1}, \qquad K^{(m)} = -q, \qquad D^{(m)} = Q$$

$m = 9j + 6$, $j \notin \{0\}$:

$$A^{(m)} = -\frac{Qq^3q^{4j} - Qq\,q^{2j}\,(1+q^n) + q^n}{(q-1)^2q^{2j}}, \qquad B^{(m)} = \frac{Q\left(2q^2q^{2j} - q^n - 1\right)}{q-1}, \qquad C^{(m)} = -Qq^{2\,j+1}$$

$$K^{(m)} = q, \qquad D^{(m)} = -q + 1$$

$m = 9j + 7$:

$$A^{(m)} = q^{-2\,j+n-4}, \qquad B^{(m)} = -\frac{Qq\,q^{2j}\,(1+q^n) - 2q^n}{q(q-1)q^{2j}}$$

$$C^{(m)} = \frac{Qq^5q^{4j} - Qq^3q^{2j}\,(1+q^n) + q^2q^n}{(q-1)^2q^{2j}}, \qquad K^{(m)} = q^{-2\,j+n-4}, \qquad D^{(m)} = -\frac{Qq^{2\,j+2} - q^n}{(q^3 - q^2)q^{2\,j}}$$

$m = 9j + 8$:

$$A^{(m)} = \frac{Qq^4(q-1)q^{4j} - Qq^2q^{2j}\,(1-q^n) - (q-1)q^n}{q^2(q-1)q^{2j}}$$

$$B^{(m)} = -\frac{Qq^2q^{2j}\,(1-q^n) + 2(q-1)q^n}{q^2(q-1)q^{2j}}, \qquad C^{(m)} = -q^{-2\,j+n-2}, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

$m = 9j + 9$:

$$A^{(m)} = -Qq^{2\,j}, \qquad B^{(m)} = \frac{Q\left(2q^2(q-1)q^{2j} + q^n - 1\right)}{q-1}$$

$$C^{(m)} = -\frac{Qq^4(q-1)q^{4j} - Qq^2q^{2j}\,(1-q^n) - (q-1)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = -q^{2\,j}$$

$$D^{(m)} = \frac{q^{2\,j+2}-1}{q-1}$$

D.2. **Pass 2.** $m = 1$, $m = 9j+1$, $j = 0$:

$$A^{(m)} = \frac{Qq^n + (2q-1)(q-2)}{8\,(q-1)}, \qquad B^{(m)} = \frac{2\,q^3 + Qq^{n+1} - 3\,q^2 + 2\,q - 2}{2\,(q-1)}$$

$$C^{(m)} = \frac{q^2(2q^2-q+2) + Q(q-2)(q+2)q^n}{2\,(q-1)}, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = \frac{3}{2}\,q + 1$$

$m = 9j+1$, $j \notin \{0\}$:

$$A^{(m)} = \frac{Qq\left(q^{2j} + (q^{2j}-1)q^n\right) - q^{4j+2}}{(q-1)^2q^{2j}}, \qquad B^{(m)} = -\frac{Q(q^n+1) - 2q^{2j+1}}{q-1}, \qquad C^{(m)} = -q^{2\,j}$$

$$K^{(m)} = q, \qquad D^{(m)} = -q + 1$$

$m = 2$, $m = 9j+2$, $j = 0$:

$$A^{(m)} = \frac{2\left(2q(q^2-q-1) + Q(q+1)q^n\right)}{q(q-1)}, \qquad B^{(m)} = -\frac{4q^3 - 7q^2 - q + 1 + Q(2q+1)q^n}{q-1}$$

$$C^{(m)} = \frac{q^2\left((2q-1)(q-2) + Qq^n\right)}{2\,(q-1)}, \qquad K^{(m)} = 4, \qquad D^{(m)} = -2\,q + 1$$

$m = 9j+2$, $j \notin \{0\}$:

$$A^{(m)} = Qq^{-2\,j+n-3}, \qquad B^{(m)} = -\frac{Q\left(q^{2j} + (q^{2j}-2)q^n\right)}{(q-1)q^{2j}}$$

$$C^{(m)} = -\frac{Qq^3\left(q^{2j} + (q^{2j}-1)q^n\right) - q^{4j+4}}{(q-1)^2q^{2j}}, \qquad K^{(m)} = q^{-2\,j+n-3}, \qquad D^{(m)} = -\frac{q^{2\,j+1} - q^n}{(q^2-q)q^{2\,j}}$$

$m = 3$, $m = 9j+3$, $j = 0$:

$$A^{(m)} = \frac{1}{2}\,Qq^{n-3}, \qquad B^{(m)} = \frac{Qq^n - Qq}{q^2-q}, \qquad C^{(m)} = -\frac{2q^2(q^2-q-1) + Qq(q+1)q^n}{2\,(q-1)}$$

$$K^{(m)} = \frac{1}{2}\,q^{n-3}, \qquad D^{(m)} = \frac{q^n - q}{q^2-q}$$

$m = 9j+3$, $j \notin \{0\}$:

$$A^{(m)} = \frac{q^2(q-1)q^{4j} + Q\left((q^{2j+1} - q + 1)q^n - q^{2j+1}\right)}{(q^2-q)q^{2j}}$$

$$B^{(m)} = -\frac{Q\left(q^{2j+1} - (q^{2j+1} - 2q + 2)q^n\right)}{(q^2-q)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-1}, \qquad K^{(m)} = 1$$

$$D^{(m)} = 1$$

$m = 4$, $m = 9j+4$, $j = 0$:

$$A^{(m)} = \frac{2q(q^2-q-1) + Q(q+1)q^n}{4\,(q^2-q)}, \qquad B^{(m)} = \frac{Qq^n - Qq}{q^2-q}, \qquad C^{(m)} = -Qq^{n-1}$$

$$K^{(m)} = \frac{1}{2}, \qquad D^{(m)} = -2\,q + 1$$

$m = 9j+4$, $j \notin \{0\}$:

$$A^{(m)} = -q^{2\,j-1}, \qquad B^{(m)} = \frac{2q(q-1)q^{2j} + Q(q^n-1)}{q-1}$$
$$C^{(m)} = -\frac{(q^4-q^3)q^{4j} - Q\left(q^{2j+2} + (q^2-q-q^{2j+2})q^n\right)}{(q-1)q^{2j}}, \qquad K^{(m)} = -q^{2\,j-1}$$
$$D^{(m)} = -\frac{Q-q^{2\,j+1}}{q-1}$$

$m = 5$, $m = 9j+5$, $j = 0$:

$$A^{(m)} = -\frac{2\left(Qq^n + (2q-1)(q-2)\right)}{q-1}, \qquad B^{(m)} = -\frac{4q^3-7q^2-q+1+Q(2q+1)q^n}{q-1}$$
$$C^{(m)} = -\frac{2q^2(q^2-q-1)+Qq(q+1)q^n}{2\,(q-1)}, \qquad K^{(m)} = 4, \qquad D^{(m)} = \frac{3}{2}\,q+1$$

$m = 9j+5$, $j \notin \{0\}$:

$$A^{(m)} = -\frac{q(q-2)^2q^{4j} + Q(q-2)q^{2j}(1+q^n) + Qq^n}{(q-1)^2q^{2j}}, \qquad B^{(m)} = -\frac{2q(q-2)q^{2j} + Q(q^n+1)}{q-1}$$
$$C^{(m)} = -q^{2\,j+1}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 9j+6$:

$$A^{(m)} = \frac{Q(q-2)\left(q^{2j+2} + (q^{2j+2}+q-2)q^n\right) + q^{4j+5}}{8q^2(q-1)^2q^{2j}}$$
$$B^{(m)} = \frac{(q^4-2q^3)q^{4j} + Q(q^2-2q+2)q^{2j}(1+q^n) + Q(q-2)q^n}{2(q-1)^2q^{2j}}$$
$$C^{(m)} = \frac{(q^5-4q^4+4q^3)q^{4j} + Qq^2(q-2)q^{2j}(1+q^n) + Qq^2q^n}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = -\frac{1}{4}$$
$$D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 9j+7$:

$$A^{(m)} = 2\,Qq^{-2\,j+n-4}, \qquad B^{(m)} = -\frac{Q\left(q^{2j+2} + (q^{2j+2}+2q-4)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$C^{(m)} = \frac{Q(q-2)\left(q^{2j+2} + (q^{2j+2}+q-2)q^n\right) + q^{4j+5}}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = 2\,q^{-2\,j+n-4}$$
$$D^{(m)} = -\frac{q^{2\,j+2}-q^n}{(q^3-q^2)q^{2\,j}}$$

$m = 9j+8$:

$$A^{(m)} = \frac{(q^5-q^4)q^{4j} + Q\left(q^{2j+2} - (q^{2j+2}+q-1)q^n\right)}{(q^3-q^2)q^{2j}}$$
$$B^{(m)} = -\frac{Q\left(q^{2j+2} - (q^{2j+2}+2q-2)q^n\right)}{(q^3-q^2)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-2}, \qquad K^{(m)} = -1$$
$$D^{(m)} = 1$$

$m = 9j+9$:

$$A^{(m)} = q^{2\,j}, \qquad B^{(m)} = -\frac{2q^2(q-1)q^{2j} - Q(q^n-1)}{q-1}$$

$$C^{(m)}=\frac{(q^5-q^4)q^{4j}+Q\left(q^{2j+2}-(q^{2j+2}+q-1)q^n\right)}{(q-1)q^{2j}},\qquad K^{(m)}=q^{2\,j},\qquad D^{(m)}=-\frac{Q-q^{2\,j+2}}{q-1}$$

D.3. **Pass 3.** $m=1$, $m=9j+1$, $j=0$:

$$A^{(m)}=-\frac{Qq(q+1)-(q^2+1)q^n}{q-1},\qquad B^{(m)}=-\frac{\left(q^3-q^2+q-2\right)q^n+Qq}{q^2-q},\qquad C^{(m)}=-q^{n-2}$$
$$K^{(m)}=q,\qquad D^{(m)}=Q$$

$m=10$, $m=9j+1$, $j=1$:

$$A^{(m)}=-\frac{Qq(q+2)+(q-2)(q^2+2)q^n}{8\,(q^2-q)},\qquad B^{(m)}=\frac{Q(q^2-2)+(q^3-2\,q^2+4\,q-2)q^n}{2\,(q-1)}$$
$$C^{(m)}=-\frac{q^2\left(Q(q-2)+(q^2-2q+2)q^n\right)}{2\,(q-1)},\qquad K^{(m)}=\frac{1}{4},\qquad D^{(m)}=-\frac{1}{2}\,q+1$$

$m=9j+1$, $j\notin\{0,1\}$:

$$A^{(m)}=2\,q^{-2\,j+n-1},\qquad B^{(m)}=-\frac{Qq^{2\,j}(1+q^n)+2q(q-2)q^n}{(q-1)q^{2\,j}}$$
$$C^{(m)}=\frac{Qq^2(q-2)q^{2\,j}(1+q^n)+Qq^{4\,j+2}+q^3(q-2)^2q^n}{2\,(q-1)^2q^{2\,j}},\qquad K^{(m)}=2\,q^{-2\,j+n-1}$$
$$D^{(m)}=-\frac{Qq^{2\,j}-q^{n+1}}{(q-1)q^{2\,j}}$$

$m=2$, $m=9j+2$, $j=0$:

$$A^{(m)}=\frac{Q^2q-\left(q^3-q^2+2\,q-1\right)q^n}{q^2-q},\qquad B^{(m)}=-\frac{Q(2q^3+1)-(2\,q^3-q^2+3\,q-1)q^n}{q-1}$$
$$C^{(m)}=\frac{Qq^3(q+1)-q^2(q^2+1)q^n}{q-1},\qquad K^{(m)}=-1,\qquad D^{(m)}=q+1$$

$m=11$, $m=9j+2$, $j=1$:

$$A^{(m)}=2\,q^{n-3},\qquad B^{(m)}=-\frac{\left(q^3-q^2+3\,q-4\right)q^n+Qq}{q^2-q}$$
$$C^{(m)}=\frac{Qq^2(q+2)+q(q-2)(q^2+2)q^n}{2\,(q-1)},\qquad K^{(m)}=2\,q^{n-3},\qquad D^{(m)}=-\frac{Qq-q^n}{q^2-q}$$

$m=9j+2$, $j\notin\{0,1\}$:

$$A^{(m)}=\frac{Q(q-1)q^{4\,j}+Qq^{2\,j+1}(1-q^n)-q^2(q-1)q^n}{(q^2-q)q^{2\,j}},\qquad B^{(m)}=-\frac{Qq^{2\,j}(1-q^n)-2q(q-1)q^n}{(q-1)q^{2\,j}}$$
$$C^{(m)}=-q^{-2\,j+n+1},\qquad K^{(m)}=-1,\qquad D^{(m)}=1$$

$m=3$, $m=9j+3$, $j=0$:

$$A^{(m)}=-q^{n-3},\qquad B^{(m)}=\frac{\left(q^3-q^2+3\,q-2\right)q^n-Qq}{q^2-q}$$
$$C^{(m)}=\frac{Q^2q^2-\left(q^4-q^3+2\,q^2-q\right)q^n}{q-1},\qquad K^{(m)}=-q^{n-3},\qquad D^{(m)}=-\frac{Qq-q^n}{q^2-q}$$

$m=12$, $m=9j+3$, $j=1$:

$$A^{(m)}=\frac{Q^2q-\big(q^3-q^2+2\,q-1\big)q^n}{q^2-q},\qquad B^{(m)}=\frac{\big(q^3-q^2+3\,q-2\big)q^n-Qq}{q^2-q}$$
$$C^{(m)}=-q^{n-1},\qquad K^{(m)}=-1,\qquad D^{(m)}=q+1$$

$m=9j+3$, $j\notin\{0,1\}$:

$$A^{(m)}=Qq^{2\,j-3},\qquad B^{(m)}=-\frac{2\,Q(q-1)q^{2\,j}+Qq(1-q^n)}{q^2-q}$$
$$C^{(m)}=\frac{Qq(q-1)q^{4\,j}+Qq^{2\,j+2}(1-q^n)-q^3(q-1)q^n}{(q-1)q^{2\,j}},\qquad K^{(m)}=q^{2\,j-3},\qquad D^{(m)}=\frac{q^{2\,j}-q}{q^2-q}$$

$m=4$, $m=9j+4$, $j=0$:

$$A^{(m)}=-\frac{Qq(q+2)+(q-2)(q^2+2)q^n}{q^2-q},\qquad B^{(m)}=-\frac{\big(q^3-q^2+3\,q-4\big)q^n+Qq}{q^2-q}$$
$$C^{(m)}=-q^{n-1},\qquad K^{(m)}=2,\qquad D^{(m)}=-\frac{1}{2}\,q+1$$

$m=13$, $m=9j+4$, $j=1$:

$$A^{(m)}=\frac{Qq(q+1)-(q^2+1)q^n}{q-1},\qquad B^{(m)}=-\frac{Q(2q^3+1)-(2\,q^3-q^2+3\,q-1)q^n}{q-1}$$
$$C^{(m)}=\frac{Q^2q^2-\big(q^4-q^3+2\,q^2-q\big)q^n}{q-1},\qquad K^{(m)}=-q,\qquad D^{(m)}=Q$$

$m=9j+4$, $j\notin\{0,1\}$:

$$A^{(m)}=-\frac{Qq^{4\,j+1}-Qq^{2\,j+1}(1+q^n)+q^2q^n}{(q-1)^2q^{2\,j}},\qquad B^{(m)}=\frac{Q\left(2q^{2\,j}-q^n-1\right)}{q-1}$$
$$C^{(m)}=-Qq^{2\,j-1},\qquad K^{(m)}=q,\qquad D^{(m)}=-q+1$$

$m=5$, $m=9j+5$, $j=0$:

$$A^{(m)}=\frac{(q^2-2\,q+2)q^n+Q(q-2)}{8\,(q-1)},\qquad B^{(m)}=\frac{Q(q^2-2)+(q^3-2\,q^2+4\,q-2)q^n}{2\,(q-1)}$$
$$C^{(m)}=\frac{Qq^2(q+2)+q(q-2)(q^2+2)q^n}{2\,(q-1)},\qquad K^{(m)}=\frac{1}{4},\qquad D^{(m)}=\frac{1}{2}\,q+1$$

$m=9j+5$, $j\notin\{0\}$:

$$A^{(m)}=q^{-2\,j+n-2},\qquad B^{(m)}=-\frac{Qq^{2\,j}(1+q^n)-2q\,q^n}{(q-1)q^{2\,j}}$$
$$C^{(m)}=\frac{Qq^{4\,j+3}-Qq^{2\,j+3}(1+q^n)+q^4q^n}{(q-1)^2q^{2\,j}},\qquad K^{(m)}=q^{-2\,j+n-2},\qquad D^{(m)}=-\frac{Qq^{2\,j}-q^n}{(q-1)q^{2\,j}}$$

$m=6$, $m=9j+6$, $j=0$:

$$A^{(m)}=2\,q^{n-2},\qquad B^{(m)}=\frac{\big(q^2-3\,q+3\big)q^n-Q}{q-1},\qquad C^{(m)}=-\frac{q^2\left(Q(q-2)+(q^2-2q+2)q^n\right)}{2\,(q-1)}$$
$$K^{(m)}=2\,q^{n-2},\qquad D^{(m)}=-\frac{Q-q^n}{q-1}$$

$m=9j+6$, $j\notin\{0\}$:

$$A^{(m)}=\frac{Q(q-1)q^{4\,j}+Qq^{2\,j}(q^n-1)-(q-1)q^n}{(q-1)q^{2\,j}},\qquad B^{(m)}=-\frac{Qq^{2\,j}(1-q^n)+2(q-1)q^n}{(q-1)q^{2\,j}}$$

$$C^{(m)} = -q^{-2\,j+n}, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

$m = 7$, $m = 9j + 7$, $j = 0$:

$$A^{(m)} = q^{n+1}, \qquad B^{(m)} = -\frac{\left(q^2 - 3\,q + 1\right)q^n + Q}{q-1}, \qquad C^{(m)} = -q^n, \qquad K^{(m)} = q^{n+1}$$

$$D^{(m)} = -\frac{\left(q^2 - 3\,q + 1\right)q^n + Q}{q-1}$$

$m = 9j + 7$, $j \notin \{0\}$:

$$A^{(m)} = -Qq^{2\,j-2}, \qquad B^{(m)} = \frac{2\,(q^3 - 2\,q^2 + 2\,q - 1)q^{2\,j} + Q(q^n - 1)}{q-1}$$

$$C^{(m)} = -\frac{Qq^2(q-1)q^{4\,j} - Qq^{2\,j+2}(1-q^n) - q^2(q-1)q^n}{(q-1)q^{2\,j}}, \qquad K^{(m)} = -q^{2\,j-2}$$

$$D^{(m)} = \frac{q^{2\,j} - 1}{q-1}$$

$m = 8$, $m = 9j + 8$, $j = 0$:

$$A^{(m)} = q^{n-2}, \qquad B^{(m)} = -\frac{\left(q^2 - 3\,q + 1\right)q^n + Q}{q-1}, \qquad C^{(m)} = -q^{n+3}, \qquad K^{(m)} = q^{n-2}$$

$$D^{(m)} = -\frac{Q - q^n}{q-1}$$

$m = 9j + 8$, $j \notin \{0\}$:

$$A^{(m)} = -\frac{Q(q-2)^2q^{4\,j} + Q(q-2)q^{2\,j}(1+q^n) + q\,q^n}{(q-1)^2q^{2\,j}}, \qquad B^{(m)} = -\frac{Q\left(2(q-2)q^{2\,j} + q^n + 1\right)}{q-1}$$

$$C^{(m)} = -Qq^{2\,j}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

$m = 9$, $m = 9j + 9$, $j = 0$:

$$A^{(m)} = \frac{(q^2 - 2\,q + 2)q^n + Q(q-2)}{q-1}, \qquad B^{(m)} = \frac{\left(q^2 - 3\,q + 3\right)q^n - Q}{q-1}, \qquad C^{(m)} = -q^n$$

$$K^{(m)} = 2, \qquad D^{(m)} = \frac{1}{2}\,q + 1$$

$m = 9j + 9$, $j \notin \{0\}$:

$$A^{(m)} = \frac{Qq(q-2)q^{2\,j}(1+q^n) + Qq^{4\,j+3} + (q-2)^2q^n}{8\,q(q-1)^2q^{2\,j}}$$

$$B^{(m)} = \frac{Qq^2(q-2)q^{4\,j} + Q(q^2 - 2q + 2)q^{2\,j}(1+q^n) + q(q-2)q^n}{2\,(q-1)^2q^{2\,j}}$$

$$C^{(m)} = \frac{Qq^2(q-2)^2q^{4\,j} + Qq^2(q-2)q^{2\,j}(1+q^n) + q^3q^n}{2\,(q-1)^2q^{2\,j}}, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q + 1$$

Appendix E. Odd $n \geq 5$: passes 1–2 and the bridge

For odd n the orbit has pre-period 1 and period $9n-13$. Passes 1 and 2 below (residues $m = 9j+r$, with the $j = 0$ and $j \in \{0, 1\}$ overrides) cover $9n - 20$ of the period tuples; the remaining 7 tuples per period form the bridge (§E.3), after which the orbit returns to pass 1 at $m = 2$. Pass 1 coincides with the even pass 1 of Appendix D. The transitions are proved for all odd n as described in §3.4, each additionally cross-checked against a direct run of Algorithm N at one concrete (n, j).

E.1. **Pass 1 (transient head).** The tuples are exactly those of even pass 1 (Appendix D, Pass 1): the same closed-form expressions in q, q^{2j} and q^n, now with n odd rather than even. We do not reproduce them.

E.2. **Pass 2.** $m = 1$, $m = 9j + 1$, $j = 0$:

$$A^{(m)} = \frac{Qq^n - q}{q-1}, \qquad B^{(m)} = -\frac{Q\big((q-2)q^n + q\big)}{q^2 - q}, \qquad C^{(m)} = -Qq^{n-2}, \qquad K^{(m)} = q$$
$$D^{(m)} = Q$$

$m = 9j + 1$, $j \notin \{0\}$:

$$A^{(m)} = \frac{(q-1)q^{4j+2} + Qq^{2j+1}(1-q^n) - Q(q-1)q^n}{(q^2-q)q^{2j}}$$
$$B^{(m)} = -\frac{Qq^{2j+1}(1-q^n) - 2Q(q-1)q^n}{(q^2-q)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-1}, \qquad K^{(m)} = -1$$
$$D^{(m)} = 1$$

$m = 2$, $m = 9j + 2$, $j = 0$:

$$A^{(m)} = -Qq^{n-1}, \qquad B^{(m)} = \frac{Q\big((2q-1)q^n - 1\big)}{q-1}, \qquad C^{(m)} = \frac{q^3 - Qq^{n+2}}{q-1}, \qquad K^{(m)} = -q^{n-1}$$
$$D^{(m)} = \frac{(2\,q-1)q^n - 1}{q-1}$$

$m = 9j + 2$, $j \notin \{0\}$:

$$A^{(m)} = q^{2\,j-1}, \qquad B^{(m)} = -\frac{2q(q-1)q^{2j} + Q(1-q^n)}{q-1}$$
$$C^{(m)} = \frac{(q-1)q^{4j+3} + Qq^{2j+2}(1-q^n) - Q(q^2-q)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = q^{2\,j-1}$$
$$D^{(m)} = -\frac{Q - q^{2\,j+1}}{q-1}$$

$m = 3$, $m = 9j + 3$, $j = 0$:

$$A^{(m)} = -\frac{Qq^n - q}{q-1}, \qquad B^{(m)} = \frac{Q\big((2q-1)q^n - 1\big)}{q-1}, \qquad C^{(m)} = -Qq^{n+1}, \qquad K^{(m)} = -q$$
$$D^{(m)} = Q$$

$m = 9j + 3$, $j \notin \{0\}$:

$$A^{(m)} = \frac{Qq^{2j+1}(1+q^n) - Qq^n - q^{4j+3}}{(q-1)^2q^{2j}}, \qquad B^{(m)} = -\frac{Q(q^n+1) - 2q^{2j+2}}{q-1}, \qquad C^{(m)} = -q^{2\,j+1}$$
$$K^{(m)} = q, \qquad D^{(m)} = -q + 1$$

$m = 9j + 4$:

$$A^{(m)} = Qq^{-2\,j+n-4}, \qquad B^{(m)} = -\frac{Q\big(q^{2j+1}(1+q^n) - 2q^n\big)}{(q^2-q)q^{2j}}$$
$$C^{(m)} = -\frac{Qq^{2j+3}(1+q^n) - Qq^2q^n - q^{4j+5}}{(q-1)^2q^{2j}}, \qquad K^{(m)} = q^{-2\,j+n-4}, \qquad D^{(m)} = -\frac{q^{2\,j+2} - q^n}{(q^3-q^2)q^{2\,j}}$$

$m = 9j + 5$:

$$A^{(m)} = \frac{(q-1)q^{4j+4} - Qq^{2j+2}(1-q^n) - Q(q-1)q^n}{(q^3-q^2)q^{2j}}$$

$$B^{(m)} = -\frac{Qq^{2j+2}(1-q^n) + 2Q(q-1)q^n}{(q^3-q^2)q^{2j}}, \qquad C^{(m)} = -Qq^{-2\,j+n-2}, \qquad K^{(m)} = 1, \qquad D^{(m)} = 1$$

$m = 9j+6$:

$$A^{(m)} = -q^{2\,j}, \qquad B^{(m)} = \frac{2q^2(q-1)q^{2j} + Q(q^n-1)}{q-1}$$

$$C^{(m)} = -\frac{(q-1)q^{4j+4} - Qq^{2j+2}(1-q^n) - Q(q-1)q^n}{(q-1)q^{2j}}, \qquad K^{(m)} = -q^{2\,j}$$

$$D^{(m)} = -\frac{Q - q^{2\,j+2}}{q-1}$$

$m = 9j+7$:

$$A^{(m)} = -\frac{(q-2)^2q^{4j+3} + Q(q-2)q^{2j+1}(1+q^n) + Qq^n}{q(q-1)^2q^{2j}}$$

$$B^{(m)} = -\frac{2q^2(q-2)q^{2j} + Q(q^n+1)}{q-1}, \qquad C^{(m)} = -q^{2\,j+2}, \qquad K^{(m)} = 2, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 9j+8$:

$$A^{(m)} = \frac{q^{4j+7} + Q(q-2)q^{2j+3}(1+q^n) + Q(q-2)^2q^n}{8q^3(q-1)^2q^{2j}}$$

$$B^{(m)} = \frac{(q-2)q^{4j+5} + Q(q^2-2q+2)q^{2j+1}(1+q^n) + Q(q-2)q^n}{2q(q-1)^2q^{2j}}$$

$$C^{(m)} = \frac{(q-2)^2q^{4j+4} + Q(q-2)q^{2j+2}(1+q^n) + Qq^{n+1}}{2(q-1)^2q^{2j}}, \qquad K^{(m)} = -\frac{1}{4}, \qquad D^{(m)} = -\frac{1}{2}\,q+1$$

$m = 9j+9$:

$$A^{(m)} = 2\,Qq^{-2\,j+n-5}, \qquad B^{(m)} = -\frac{Q\big(q^{2j+3}(1+q^n) + 2(q-2)q^n\big)}{(q^4-q^3)q^{2j}}$$

$$C^{(m)} = \frac{q^{4j+7} + Q(q-2)q^{2j+3}(1+q^n) + Q(q-2)^2q^n}{2q(q-1)^2q^{2j}}, \qquad K^{(m)} = 2\,q^{-2\,j+n-5}$$

$$D^{(m)} = -\frac{q^{2\,j+3} - q^n}{(q^4-q^3)q^{2\,j}}$$

E.3. **The 7-tuple bridge (odd n).** For odd $n \geq 5$, the seven tuples following the pass-2 boundary are:

$$A^{(\mathrm{b1})} = \frac{qQ^2 - (q^3-q^2+2q-1)q^n}{q^2-q}, \qquad B^{(\mathrm{b1})} = \frac{(2q^3-q^2+3q-1)q^n - Q(2q^3+1)}{q-1}$$

$$C^{(\mathrm{b1})} = \frac{q^2\big(q^4+q-(q^2+1)q^n\big)}{q-1}, \qquad K^{(\mathrm{b1})} = -1, \qquad D^{(\mathrm{b1})} = q+1.$$

$$A^{(\mathrm{b2})} = -q^{n-3}, \qquad B^{(\mathrm{b2})} = \frac{(q^3-q^2+3q-2)q^n - qQ}{q^2-q}$$

$$C^{(\mathrm{b2})} = \frac{q\big(qQ^2 - (q^3-q^2+2q-1)q^n\big)}{q-1}, \qquad K^{(\mathrm{b2})} = -q^{n-3}, \qquad D^{(\mathrm{b2})} = \frac{q^n - qQ}{q^2-q}.$$

$$A^{(\mathrm{b3})}=-\frac{q(q+2)Q+(q-2)(q^2+2)q^n}{q^2-q},\qquad B^{(\mathrm{b3})}=-\frac{qQ+(q^3-q^2+3q-4)q^n}{q^2-q}$$
$$C^{(\mathrm{b3})}=-q^{n-1},\qquad K^{(\mathrm{b3})}=2,\qquad D^{(\mathrm{b3})}=-\tfrac{1}{2}q+1.$$

$$A^{(\mathrm{b4})}=\frac{(q-2)Q+(q^2-2q+2)q^n}{8(q-1)},\qquad B^{(\mathrm{b4})}=\frac{(q^2-2)Q+(q^3-2q^2+4q-2)q^n}{2(q-1)}$$
$$C^{(\mathrm{b4})}=\frac{q\big(q(q+2)Q+(q-2)(q^2+2)q^n\big)}{2(q-1)},\qquad K^{(\mathrm{b4})}=\tfrac{1}{4},\qquad D^{(\mathrm{b4})}=\tfrac{1}{2}q+1.$$

$$A^{(\mathrm{b5})}=2q^{n-2},\qquad B^{(\mathrm{b5})}=\frac{(q^2-3q+3)q^n-Q}{q-1}$$
$$C^{(\mathrm{b5})}=-\frac{q^2\big((q-2)Q+(q^2-2q+2)q^n\big)}{2(q-1)},\qquad K^{(\mathrm{b5})}=2q^{n-2},\qquad D^{(\mathrm{b5})}=\frac{q^n-Q}{q-1}.$$

$$A^{(\mathrm{b6})}=q^{n+1},\qquad B^{(\mathrm{b6})}=-\frac{(q^2-3q+1)q^n+Q}{q-1},\qquad C^{(\mathrm{b6})}=-q^n$$
$$K^{(\mathrm{b6})}=q^{n+1},\qquad D^{(\mathrm{b6})}=-\frac{(q^2-3q+1)q^n+Q}{q-1}.$$

$$A^{(\mathrm{b7})}=q^{n-2},\qquad B^{(\mathrm{b7})}=-\frac{(q^2-3q+1)q^n+Q}{q-1},\qquad C^{(\mathrm{b7})}=-q^{n+3}$$
$$K^{(\mathrm{b7})}=q^{n-2},\qquad D^{(\mathrm{b7})}=\frac{q^n-Q}{q-1}.$$

Acknowledgements

The author thanks Emmanuel Pedon for valuable discussions.

References

[1] G.-N. Han, *Hankel continued fraction and its applications*, Adv. Math. **303** (2016), 295–321.
[2] G.-N. Han and E. Pedon, *Hankel continued fractions and Hankel determinants for q-deformed metallic numbers*, arXiv:2502.05993, 2025.
[3] S. Morier-Genoud and V. Ovsienko, *On q-deformed real numbers*, Exp. Math. **31** (2022), no. 2, 652–660.
[4] V. Ovsienko and E. Pedon, *Continued fractions for q-deformed real numbers, $\{-1,0,1\}$-Hankel determinants, and Somos–Gale–Robinson sequences*, Adv. in Appl. Math. **162** (2025), Paper No. 102788, 37 pp.

Université de Strasbourg, CNRS, IRMA, UMR 7501, Strasbourg, France
Email address: guoniu.han@unistra.fr